

# WASP-36b

R-2422

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-36\_130207 · created 2026-08-21T23:47:44+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                  |
|---------------------------------------------------|----------------------------------|
| Mid-Transit Time (Tmid)                           | 2456330.8347 +/- 0.00049 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.09 +/- 0.032                   |
| Transit depth (Rp/Rs)^2                           | 0.81 +/- 0.57 [%]                |
| Orbital Inclination (inc)                         | 88.11 +/- 2.92                   |
| Airmass coefficient 1 (a1)                        | 1.037 +/- 0.015                  |
| Airmass coefficient 2 (a2)                        | -0.026 +/- 0.011                 |
| Scatter in the residuals of the lightcurve fit is | 1.47 %                           |
| Best Comparison Star                              | #1 - [591, 80]                   |
| Optimal Aperture                                  | 3.41                             |
| Optimal Annulus                                   | 13.13                            |
| Transit Duration (day)                            | 0.0869 +/- 0.0084                |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                          |
|---------------------------|------------------------------------------|
| Rp/R■ fit vs published    | 0.09 ± 0.032 vs 0.1368 (deviation 1.33σ) |
| Duration fit vs published | 0.0869 d vs 0.0758 d (deviation 1.2σ)    |
| Mid-time O-C              | 1.2 min                                  |
| Depth SNR                 | 2.6                                      |
| Residual scatter          | 1.47 %                                   |

### Light curve

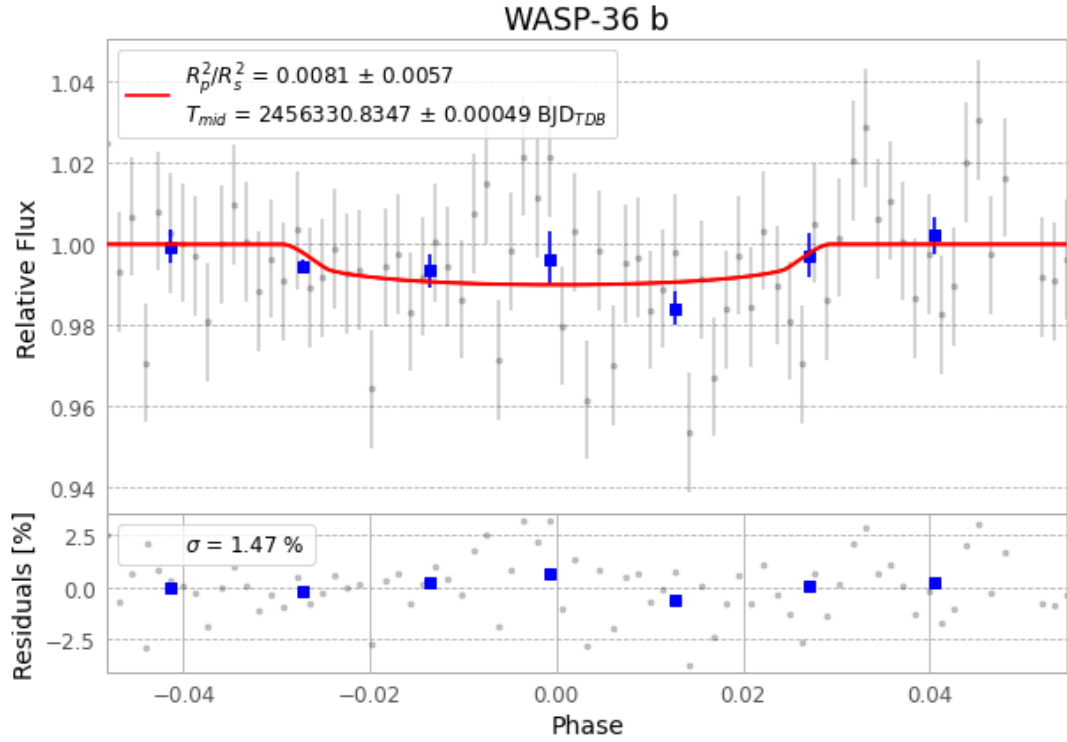


Figure 1 — FinalLightCurve\_WASP-36 b\_2013-02-06.png (SHA-256 1460bd2ded0017a7...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [280, 306], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 3dcda60fe886501d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit a4253ce

## Data provenance

77 FITS frames (51 MB), WASP-36130207060612.FITS → WASP-36130207095412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-130207.log (b34e0482621f...), FinalLightCurve\_WASP-36 b\_2013-02-06.png (1460bd2ded00...), FinalLightCurve\_WASP-36 b\_2013-02-06.csv (138e25ae7db9...)

## Compute resources

Wall time 107.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

---

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-21 23:47Z.